Who are ACDC's members?

Exploratory data analysis for customer segmentation

Data Mining, NOVA IMS | Group 10 | Deliverable 1: Exploratory Data Analysis

A report on what the membership data already says about ACDC's customer base, what it cannot say yet, and what both of those imply for the segmentation that follows.

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What ACDC handed us, and what the file needed before it could be read

ACDC wants member groups it can market to. The raw file is one row per member - plan, tenure and three years of space usage - and it is not analysis-ready. Every number in this report comes from the prepared table, where each repair stays visible in its own flag column.

One variable owned the geometry - On raw values, lifetime value accounts for 99.9% of a typical distance between two members: a distance-based method would have compared revenue and nothing else.

Repairs, not deletions - 15 exact duplicate rows were removed; 3,238 spelling variants and 637 day-first dates were repaired, and 1,742 zero-value placeholders became missing - all of them flagged.

Missing stays missing - 18.5% of lifetime value is genuinely absent, and roughly nine per cent of the recency, meeting-room and guest-pass columns are too; that is carried as missing, never filled with a plausible number.

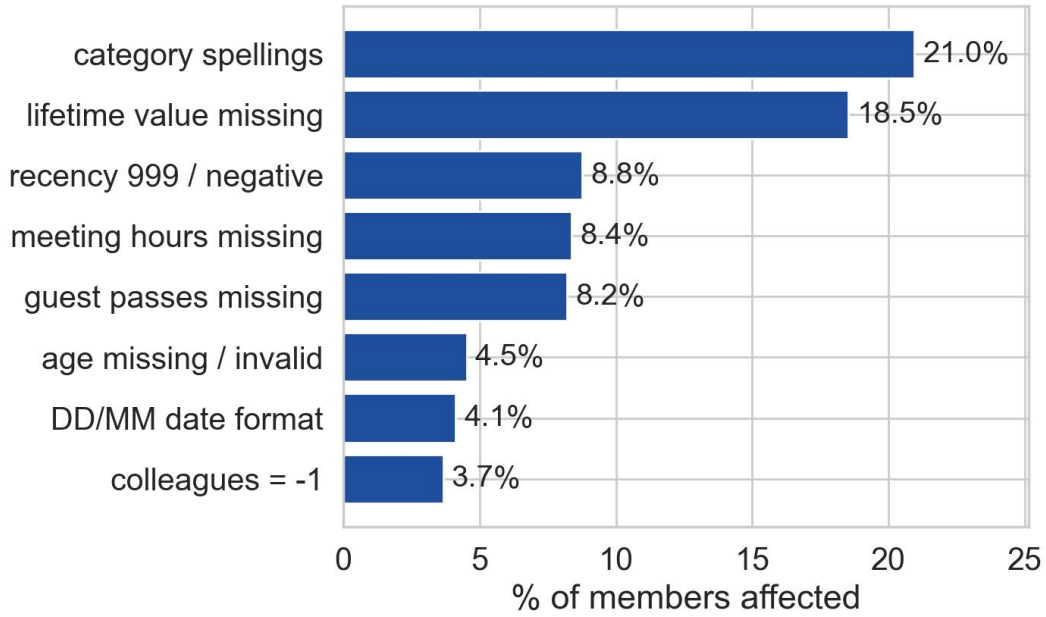


Figure 1. *Share of members affected by each issue, computed from the flags and sorted for reading.*

What this means for the segmentation - Preparation decides what a distance can see, so the cleaning log and 52 flag columns travel with the data: any finding here traces back to the cell that produced it.

A few members carry the value: median €1,668, the richest percentile opens at €26,125

Lifetime value is not a bell curve, it is a tail. The mean sits more than four times above the median, and the richest one per cent of members alone holds 46% of all known value.

The zeros are placeholders - 1,742 members report no value at all. Read as spending they pull the median down to €859; they are spread evenly across plans, so they are treated as missing.

The extremes are members - 17 members sit beyond the €1M fence. They were kept and shown on a log scale: an extreme is a fact about the business, not a mistake to delete.

Rank holds, raw distance does not - Usage and value move together in rank (0.77) but barely in euros (0.15). Groups should be measured on ranks or logs, never on raw amounts.

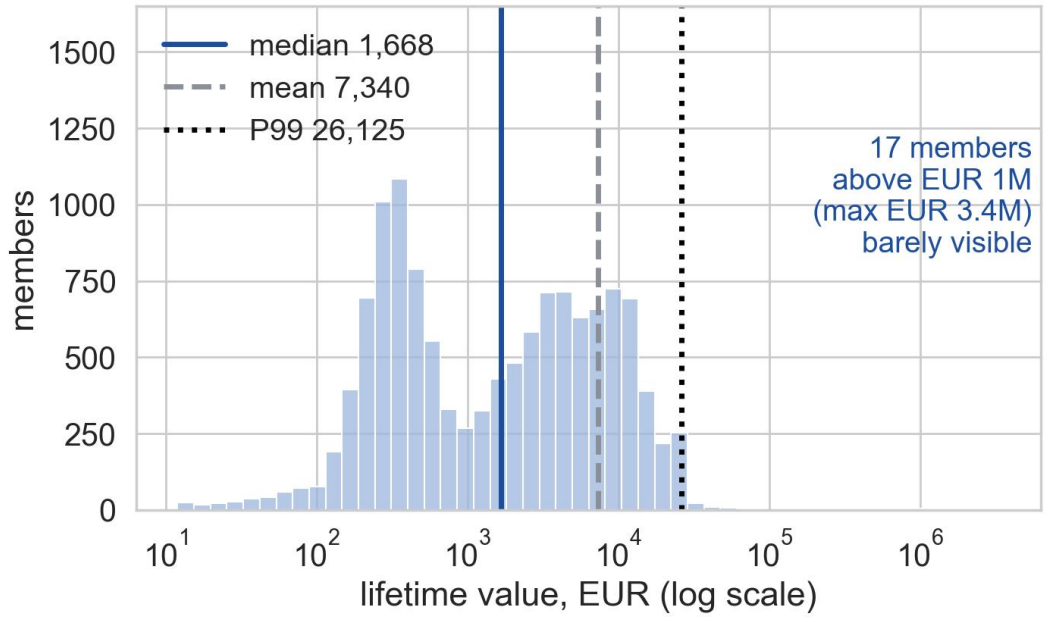


Figure 2. *The same distribution on a log axis: the median, mean and richest percentile are far apart.*

What this means for the segmentation - A value-based segment will be small and rich. Expect one group to hold most of the revenue, and expect its boundaries to move whenever a zero or an extreme is reclassified.

The plans do split revenue - just not in the order they are sold

The dataset annex presents the plans as an order of commitment, and the entitled days per week do rise with it. Median revenue per month rises too, with one exception that no cleaning step can explain.

Five real earning levels - All 4 pairs of neighbouring plans have non-overlapping interquartile ranges on monthly value, so the plans separate the base into 5 genuine earning levels.

One rung is upside down - Full-time members realise €313 a month against €408 for dedicated-desk members, so the commitment ladder and the revenue ladder disagree on this pair.

A label is not a price - The plan scale is ordinal: it carries rank only. Nothing here averages a tier name or assumes the step between two tiers is the same size as the step between any other two.

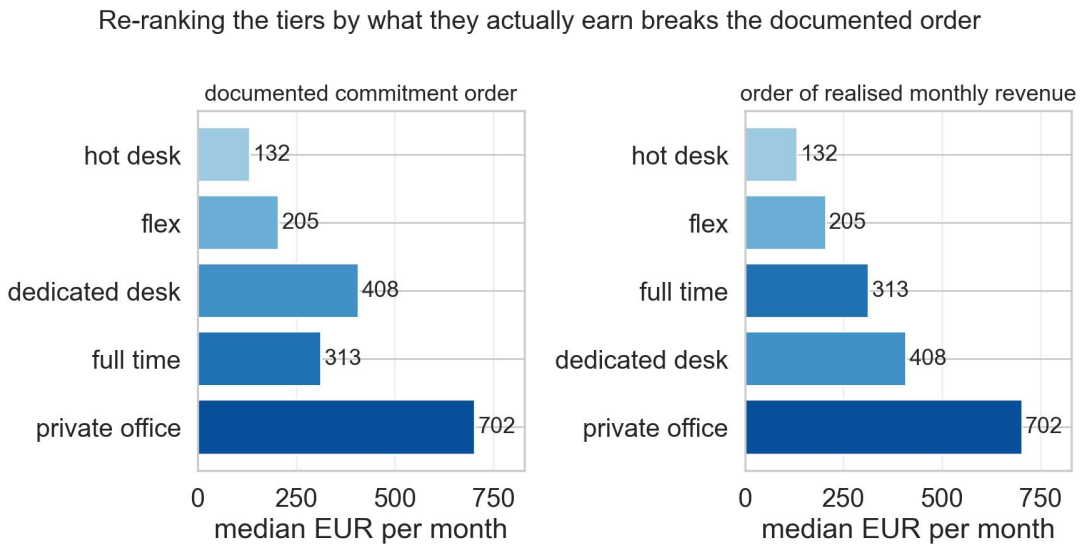


Figure 3. *Left: tiers as documented. Right: the same tiers sorted by what they actually earn.*

What this means for the segmentation - Price will pull a cluster one way and behaviour another. We would use the plan to name and read a group, not to define it - and put the inverted rung to ACDC as a pricing question.

Retention, not the price list, marks the natural groups

Every plan is mostly churned, but how many members are still active differs sharply: 57% on dedicated desks against 13% on hot desks. Usage tells us more than entitlement does.

Dormant payers - 1,112 active dedicated-desk and full-time members - 62% of that group - use under 10% of the visits their plan entitles them to.

Absent, not new - Their median gap since a visit is 195 days, against 8 days for other active members. They hold a desk; they are not in it.

Status and usage must be read together - Members who have already ended used 67% of their allowance while the active desk base uses 5% - a heavy user who left and a light user who stayed are different people.

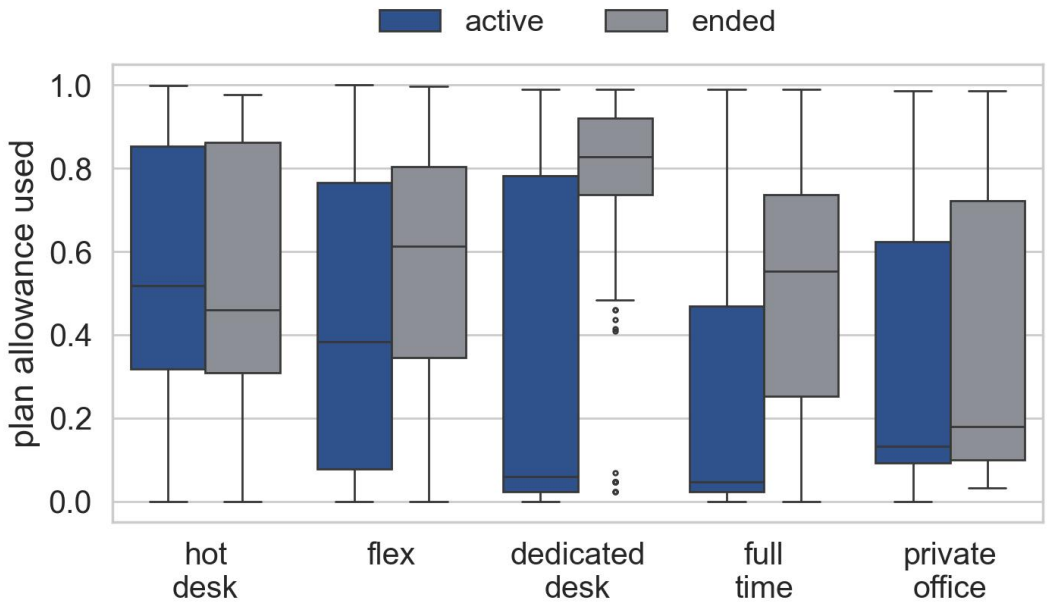


Figure 4. *Allowance used by plan and status: the active low-usage block is the finding, not the outliers.*

What this means for the segmentation - Engagement cuts the base more cleanly than price does, so the segmentation should be driven by usage and recency features with the plan kept for interpretation.

For ACDC: expect 4-6 groups, told apart by behaviour rather than by the price list

The plans give 5 clean revenue levels; usage adds a dormant-versus-engaged split and a small premium tail. So we expect 4-6 groups. No clustering has been run yet: this is a descriptive judgement for the next deliverable to test.

Greatest risk to the clustering - Pooling 11,496 ended members with 3,954 active ones compares histories rather than current behaviour: the first thing a distance sees would be status.

Anomalies are not noise - 155 members in the top percentile of the anomaly score hold 21% of known revenue and more than a third are still active. They should become a described group, not a filter.

What could still change the answer - 28.8% of members match none of the descriptive rules we could state, and 18.5% have no known value, so any first grouping must be validated on behaviour.

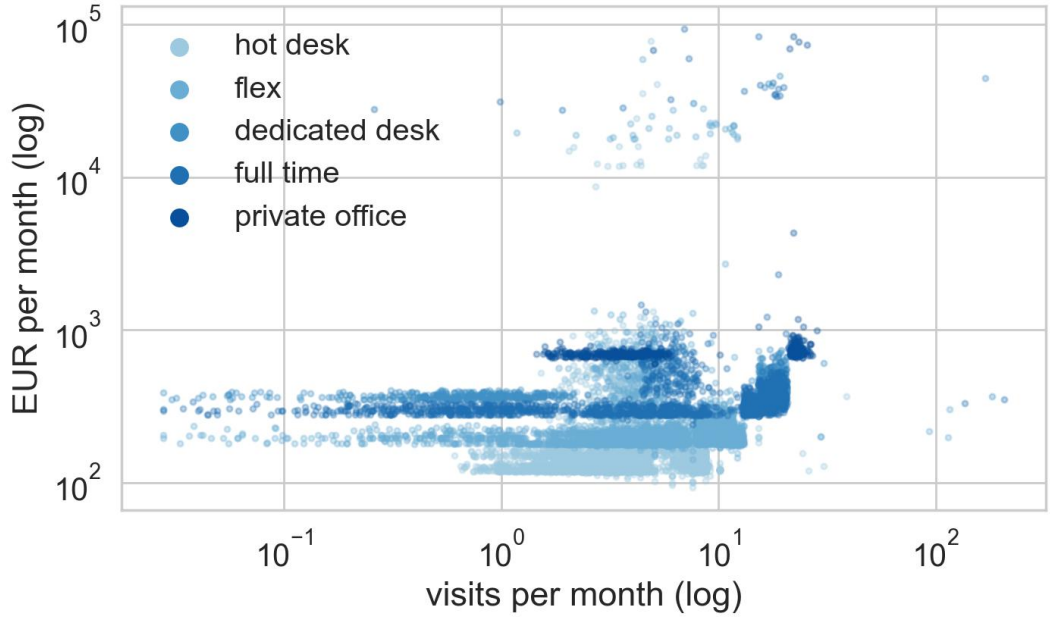


Figure 5. *Visits per month against euros per month, both on log scales, coloured by plan: the plans overlap in behaviour.*

What this means for the segmentation - Next: cluster the active base on the rescaled usage features, keep the ended population as a churn contrast, then profile each group on monthly earnings, value per visit and how many statistical fences a member breaks - in the data ltv_per_month, value_per_visit, n_iqr_fences - the three features that separate the documented populations furthest.